

Covert Routing in Heterogeneous Networks

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Abstract—In this paper, we explore covert routing communication in a heterogeneous network where a source sends a confidential message to a destination node with the help of relaying nodes where each node adaptively selects one modality among multiple communication modalities based on the wireless environment. We study three optimization problems: (i) the maximization of the end-to-end detection error probability at an adversary with a requirement on the throughput, (ii) the end-to-end throughput maximization under a covertness constraint, and (iii) the end-to-end latency minimization with a covertness condition. For the three optimization problems, we develop novel algorithms that identify the routes from a source to a destination and allocate resources, which are communication modality, transmit power, and bandwidth, at all nodes along the route. First, for single-hop communications, we derive a closed-form joint optimal power and bandwidth solution for a given modality, and then provide a modality selection method. For multi-hop communications, we propose the optimal routing strategies for the three problems by modeling the network as graphs and defining edge weights based on the objectives of the problems. From numerical simulations, it is validated that the performance of the network can be enhanced with the proposed optimal joint route and resource allocation techniques by judiciously selecting one of the multiple modalities for each hop in the route based on the wireless environment.

Index Terms—Covert communications, heterogeneous network, routing, joint optimization.

I. INTRODUCTION

Due to the open nature of wireless transmissions and the ever-increasing threats from adversaries in wireless networks, physical layer security has attracted a lot of attention over the past decades [2], [3]. There are two security protection approaches [4]: (i) Preventing adversaries from intercepting confidential messages which is called *wiretap* channel problem [5], and (ii) Preventing adversaries from detecting the existence of communications which is termed *covert* or *low probability of detection communication* [6]. Since the wiretap channel problem only protects messages, it may not be sufficient to guarantee strong security. Hence, in security-sensitive applications, it is critical to establish covert communication links to hide the presence of communications.

A. Related Work and Motivation

For the wiretap channel, the information-theoretic studies for a single-user channel, a broadcast channel, and a fading broadcast channel were investigated in [5], [7] and [8], respectively. The authors in [9] assessed the secrecy performance

of wireless networks with friendly jammers by leveraging stochastic geometry. The secrecy performance of a distributed beamforming technique with artificial interference was characterized in [10]. In addition, the work in [11] introduced distributed beamforming schemes which create a secure communication link based on two-bit feedback from a receiver without any information about potential adversaries.

For covert communication, the information-theoretic limits for an additive white Gaussian noise (AWGN) channel, a binary symmetric channel, a discrete memoryless channel, and multiple access channel were analyzed in [12]–[15], respectively. The average covert probability and covert outage probability were examined in [16] when noise uncertainty at a warden is taken into account. The studies in [17] and [18] proposed transmit power optimization strategies for full-duplex relaying systems and networks with a full-duplex jamming receiver, respectively. Also, the covert rate improvement from leveraging an intelligent reflecting surface was explored in [19] and [20]. For covert communication with finite block length, the authors in [21] developed channel inversion power control techniques for the case where a transmitter is allowed to communicate with a receiver without sending pilot signals.

When a source and a destination are far away, the source can send messages to the destination with the help of relaying nodes. For the wiretap channel, a joint routing and transmit power optimization for the end-to-end connection outage minimization under a constraint on the secrecy outage probability was introduced in [22]. The work in [23] investigated the throughput of covert communications against an unmanned aerial vehicle when the coding rates, transmit power, and number of relaying nodes, that are equidistantly placed on the line from a source to a destination, are numerically optimized. The algorithms for the joint optimization of the transmit power and path for the maximum throughput and minimum delay were provided in [24] based on a bound on the detection error probability (DEP) and an approximated covert throughput.

Recently, many research efforts have been devoted to heterogeneous networks (HetNets) where multiple independent networks are exploited to provide enhanced capacity and security as well as wider more persistent coverage [25]. In [26], the wiretap secrecy rate maximizing spectrum allocation schemes were devised for a two-tier downlink HetNet with one macrocell and several femtocells. The wiretap secrecy and connection probabilities of a downlink multi-tier HetNet were assessed in [27] utilizing a truncated average received signal power user association policy. In [28], a resource allocation method that maximizes the wiretap secrecy energy efficiency subject to the average packet delay was developed for multi-homing HetNets. In addition, for covert communication in an air-to-ground network with omnidirectional microwave and

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directional millimeter wave transmission modes, numerical solutions for transmit power optimization and mode selection were introduced in [29]. Overall, previous works in [22]–[29] either assumed a single communication modality, or only single-hop communication.

Conventional HetNets having multiple independent networks often use optimization among the different networks based on metrics such as signal strength, and mainly focus on single-hop adaptation as alluded to earlier (e.g., cellular/wireless local-area network (WLAN) modality switching on personal devices). In order to realize more resilient and secure networking capabilities, the unique and complimentary characteristics of state-of-the-art and emerging communication technologies could be exploited for multi-hop multi-modal routing to achieve end-to-end security and enhance communications performance. To the best of the authors' knowledge, routing algorithms for covert multi-hop multi-modal communication have not been reported.

In this paper, we provide novel joint route and resource optimization strategies for HetNets that are more tightly integrated to enhance covertness of the overall route while ensuring other communication performance metrics are met. Specifically, unlike conventional HetNets, we envision highly adaptive and intelligent HetNets where multi-hop multi-modal routes are dynamically established where each hop in a route adapts to the most optimal available modality.

B. Contributions and Organization

In our earlier work in [1], only the single-hop DEP maximization optimization was studied. In this study, we extend the work by considering not only DEP maximization but also throughput maximization and latency minimization for single-hop communications. Furthermore, we investigate multi-hop covert routing and propose the optimal algorithms for the three end-to-end performance optimization problems. Specifically, we consider joint route and resource allocation optimizations for covert communication in a HetNet where a source (Alice) is communicating with a destination (Bob) with the aid of relaying nodes where each node judiciously chooses one modality among multiple communication modalities based on the wireless environment, while an adversary (Willie) is attempting to detect the existence of the communication. This is the first work that investigates covert routing communication in HetNets. The contributions of this paper are detailed as follows:

- Since various communication modalities have drastically different characteristics in terms of the communication range, channel fading, data rate, directionality, etc., due to the difference in their center frequencies, it is intuitive that these varying characteristics should be taken into account when developing routing strategies for the HetNets. For this, we formulate novel joint route and resource allocation problems that optimize three performance metrics, the end-to-end DEP, end-to-end throughput, and end-to-end latency, for covert routing communication with multiple communication modalities.

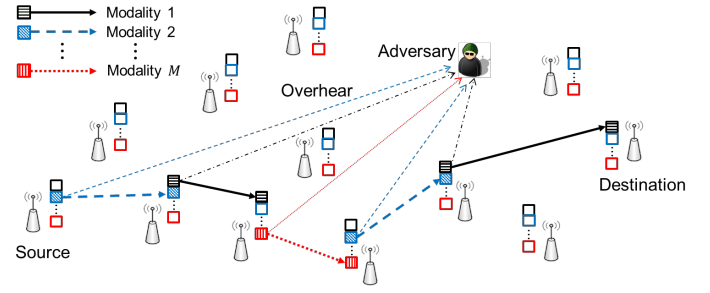


Fig. 1. A network model composed of nodes equipped with multiple communication modalities. The square colored boxes signify the different communication modalities where the squares filled with lines represent the selected modality for each hop in the example route.

- For any given route from Alice to Bob, we decompose joint multi-hop resource allocation problem into single-hop resource allocation problems. For each of single-hop optimizations, we derive a closed-form optimal joint transmit power and bandwidth optimization for a given modality. Then, we introduce methods to select the best modality for each of the performance metrics.
- For multi-hop route optimizations, we model the HetNet as directed graphs by properly defining edge weights based on the objectives of the three problems. Then, we identify the optimal routes for the three problems by formulating the routing problems as the shortest path problem, or widest path problem.
- To assess the performance of the proposed algorithms, we model the propagation in a realistic 3D environment and obtain channel data for multiple communication modalities at different frequencies using a high-fidelity ray-tracing technique [30]. From extensive numerical simulations, it is validated that the proposed optimal joint route and resource allocation strategies can enhance the end-to-end performances by adaptively choosing the optimal modality for each hop in the route among the multiple modalities available at each hop.

The organization of this paper is as follows: Section II describes the network model and formulates joint route and resource allocation optimization problems with three performance metrics for covert communication in HetNets. In Section III, we present new joint transmit power, bandwidth and modality optimization techniques for a single-hop communication. The routing solutions for the three multi-hop multi-modal routing problems are introduced in Section IV. Section V demonstrates the numerical results followed by conclusions in Section VI. All notations used in this paper are summarized in Table I.

II. NETWORK MODEL AND PROBLEM FORMULATION

A. Network Model

We examine a HetNet consisting of Alice, Bob, multiple relaying nodes, and Willie as illustrated in Fig. 1. We assume all nodes in the network including Alice, Bob, and the relay nodes are equipped with the same set of communication technologies or modalities (e.g., having different operating frequency, bandwidth, and channel characteristics). Willie is

TABLE I
LIST OF NOTATIONS

Notation	Definition
Ψ	Set of all possible routes from Alice to Bob
$\psi = (h_1, \dots, h_{N_\psi})$	A route from Alice to Bob
$h_i = (T_{h_i}, R_{h_i})$	Single-hop link between T_{h_i} and R_{h_i}
N_ψ	Number of hops of the route ψ
M	Number of communication modalities
L	Number of channel uses in a communication slot
$g_{i,j}^{(m)}$	Channel from i to j for modality m
$g_{i,W}^{(m)}$	Channel from i to Willie for modality m
$P_i^{(m)}$	Transmit power at i for modality m
$\Omega_i^{(m)}$	Allocated Bandwidth at i for modality m
N_0	Power spectral density of AWGN
δ_h	Detection threshold at Willie for link h
$P_{\text{DEP},h}$	DEP at Willie for link h
U_h	Throughput of link h
L_h	Latency of link h
$P_{\text{DEP}}(\psi)$	End-to-end DEP of route ψ
$U(\psi)$	End-to-end Throughput of route ψ
$L(\psi)$	End-to-end latency of route ψ
D	Size of the transmitted data
U_{target}	Target throughput
$P_{\text{DEP,target}}$	Target DEP

a passive adversary who is also equipped with a detector that covers the frequency bands being used by all modalities in the network. Alice transmits a confidential message to Bob with the help of relaying nodes while Willie is trying to detect the presence of communication from Alice to Bob. We define Ψ as the set of all possible routes from Alice to Bob. Let us denote $\psi = (h_1, \dots, h_{N_\psi}) \in \Psi$ as a route from Alice to Bob where $h_i = (T_{h_i}, R_{h_i})$ is the single-hop communication link between transmitter T_{h_i} and receiver R_{h_i} , and N_ψ is the number of hops of the route ψ . Here, T_{h_1} and $R_{h_{N_\psi}}$ are Alice and Bob, respectively. In this network, each of the transmitters $\{T_{h_i}\}$ along the path ψ adaptively selects one of M communication modalities based on the dynamic wireless environment and the communication requirements.

A communication slot for single-hop communication between a transmitter/receiver pair consists of a block of L channel uses, and the transmitter decides whether to send confidential data to the receiver or not with equal *a priori* probability. Then, for communication link h , when the transmitter T_h sends the data symbol $x[l] \sim \mathcal{C}(0, 1)$ for the l -th channel use by selecting modality m_h from multiple modalities, the received signals at receiver R_h and Willie are respectively given by

$$y_{T_h, R_h}^{(m_h)}[l] = \sqrt{P_{T_h}^{(m_h)}} g_{T_h, R_h}^{(m_h)} x[l] + n_{T_h, R_h}^{(m_h)}[l], \quad (1)$$

$$y_{T_h, W}^{(m_h)}[l] = \sqrt{P_{T_h}^{(m_h)}} g_{T_h, W}^{(m_h)} x[l] + n_{T_h, W}^{(m_h)}[l], \quad (2)$$

for $l = 1, \dots, L$, where $g_{T_h, R_h}^{(m_h)}$ and $g_{T_h, W}^{(m_h)}$ mean the channels from transmitter T_h to receiver R_h and Willie for modality m_h , respectively. $P_{T_h}^{(m_h)}$ stands for the transmit power at transmitter T_h for modality m_h . Also, $n_{T_h, R_h}^{(m_h)}[l] \sim \mathcal{CN}(0, \Omega_{T_h}^{(m_h)} N_0)$ and $n_{T_h, W}^{(m_h)}[l] \sim \mathcal{CN}(0, \Omega_{T_h}^{(m_h)} N_0)$ indicate the AGWN at receiver R_h and Willie for modality m_h , respectively. Here, $\Omega_{T_h}^{(m_h)}$ and N_0 specify the allocated bandwidth at transmit-

ter T_h for modality m_h and the power spectral density, respectively. In this paper, we assume that the channels are independent with each other.

B. Detection at Willie

In this paper, we assume that Willie has knowledge of the optimized route from Alice to Bob $\psi = (h_1, \dots, h_{N_\psi})$, selected modalities $\{m_h\}_{h \in \psi}$, channels from transmitters to Willie $\{g_{T_h, W}^{(m_h)}\}_{h \in \psi}$, transmit powers $\{P_{T_h}^{(m_h)}\}_{h \in \psi}$, and allocated bandwidths $\{\Omega_{T_h}^{(m_h)}\}_{h \in \psi}$. Note that this case corresponds to the worst-case scenario from a covertness perspective and is the best case for Willie.

Willie tries to detect the existence of the transmissions along the route ψ by attempting to detect each of the hops in the route. For transmission of a single-hop link $h \in \psi$, Willie attempts to distinguish the following two hypotheses:

$$\mathcal{H}_h^{(0)} : y_{T_h, W}^{(m_h)}[l] = n_{T_h, W}^{(m_h)}[l], \quad (3)$$

$$\mathcal{H}_h^{(1)} : y_{T_h, W}^{(m_h)}[l] = \sqrt{P_{T_h}^{(m_h)}} g_{T_h, W}^{(m_h)} x[l] + n_{T_h, W}^{(m_h)}[l],$$

for $l = 1, \dots, L$, where $\mathcal{H}_h^{(0)}$ signifies the null hypothesis in which there is no transmission and $\mathcal{H}_h^{(1)}$ designates the alternative hypothesis in which there exists a transmission.

To detect the presence of communication over link h , Willie measures the received signal strengths for L channel uses and computes the average received signal strength as

$$\bar{y}_{T_h, W}^{(m_h)} = \frac{1}{L} \sum_{n=1}^L \left| y_{T_h, W}^{(m_h)}[l] \right|^2. \quad (4)$$

Then, Willie makes a binary decision about whether there is communication over link h by employing a radiometer using his knowledge of the modality being used for link h (i.e., center frequency and bandwidth) and conducting a threshold test as follows [12], [13], [16], [19], [21], [31], [32]:

$$\bar{y}_{T_h, W}^{(m_h)} \underset{\mathcal{D}_h^{(0)}}{\overset{\mathcal{D}_h^{(1)}}{\gtrless}} \delta_h, \quad (5)$$

where δ_h denotes the detection threshold for link h , and $\mathcal{D}_h^{(0)}$ and $\mathcal{D}_h^{(1)}$ mean the decisions in favor of $\mathcal{H}_h^{(0)}$ and $\mathcal{H}_h^{(1)}$, respectively.

Since a transmitter decides whether to send a message or not with equal *a priori* probability, without loss of generality, the DEP at Willie for link h is written as

$$P_{\text{DEP},h} = P_{\text{MD},h} + P_{\text{FA},h}, \quad (6)$$

where $P_{\text{MD},h} = \mathbb{P}(\mathcal{D}_h^{(0)} | \mathcal{H}_h^{(1)})$ and $P_{\text{FA},h} = \mathbb{P}(\mathcal{D}_h^{(1)} | \mathcal{H}_h^{(0)})$ stand for the missed detection probability and false alarm probability, respectively. As $\bar{y}_{T_h, W}^{(m_h)}$ in (4) is a chi-squared random variable with $2L$ degrees of freedom, we can obtain the DEP in (6) as [32]

$$P_{\text{DEP},h} = 1 - \frac{\gamma \left(L, \frac{L\delta_h}{\Omega_{T_h}^{(m_h)} N_0} \right) - \gamma \left(L, \frac{L\delta_h}{P_{T_h}^{(m_h)} |g_{T_h, W}^{(m_h)}|^2 + \Omega_{T_h}^{(m_h)} N_0} \right)}{\Gamma(L)}, \quad (7)$$

where $\Gamma(s) = \int_0^\infty t^{s-1} e^{-t} dt$ and $\gamma(s, x) = \int_0^x t^{s-1} e^{-t} dt$ indicate the gamma function and lower incomplete gamma function, respectively.

C. Performance Metrics

Now, let us introduce the end-to-end performance metrics for a route $\psi \in \Psi$ from Alice to Bob. The covertness of communication over link h is measured by the DEP in (7). If Willie attempts to detect the presence of communication of a route from Alice and Bob, Willie fails to detect the communication when Willie makes wrong decisions for all hops within the route. Thus, we can represent the end-to-end DEP of route ψ as

$$\mathbb{P}_{\text{DEP}}(\psi) = \prod_{h \in \psi} \mathbb{P}_{\text{DEP},h}. \quad (8)$$

Next, for a single-hop link h between transmitter T_h to receiver R_h , the throughput of link h is expressed as

$$U_h = \Omega_{T_h}^{(m_h)} \log_2 \left(1 + \frac{P_{T_h}^{(m_h)} |g_{T_h, R_h}^{(m_h)}|^2}{\Omega_{T_h}^{(m_h)} N_0} \right). \quad (9)$$

For an end-to-end route $\psi = (h_1, \dots, h_{N_\psi})$, the single-hop link with the smallest throughput becomes the bottleneck of the entire route ψ . Therefore, we can write the end-to-end throughput of route ψ as

$$U(\psi) = \min_{h \in \psi} U_h. \quad (10)$$

Lastly, we define the communication latency of link h , which is the inverse of the link throughput, as

$$L_h = \frac{D}{U_h}, \quad (11)$$

where D is the size of data and the end-to-end latency of route ψ is given by

$$L(\psi) = \sum_{h \in \psi} L_h. \quad (12)$$

D. Problem Formulation

In this subsection, we introduce three optimization problems with various objectives. For each of the optimization problems, our goal is to jointly identify the optimal route from Alice to Bob, and the optimal modality/power/bandwidth allocation strategy for all hops along the route.

1) *End-to-end DEP Maximization*: The first optimization problem is the *end-to-end DEP maximization* with a requirement on the end-to-end throughput which is represented as

$$\max_{\psi \in \Psi, \{m_h, P_{T_h}^{(m_h)}, \Omega_{T_h}^{(m_h)}\}_{h \in \psi}} \mathbb{P}_{\text{DEP}}(\psi) \quad (13)$$

$$\text{s.t. } P_{T_h}^{(m_h)} \leq P_{\max}, \quad \forall h \in \psi, \quad (14)$$

$$\Omega_{\text{lb}}^{(m_h)} \leq \Omega_{T_h}^{(m_h)} \leq \Omega_{\text{ub}}^{(m_h)}, \quad \forall h \in \psi, \quad (15)$$

$$U(\psi) \geq U_{\text{target}}, \quad (16)$$

where $P_{\max}$ specifies the maximum available power budget at all transmitters, and $\Omega_{\text{lb}}^{(m)}$ and $\Omega_{\text{ub}}^{(m)}$ respectively denote

the lower and upper bounds of the available bandwidth for modality m . Here, U_{target} signifies the minimum required end-to-end throughput from Alice at Bob.

2) *End-to-end Throughput Maximization*: Next, we formulate the *end-to-end throughput maximization* under constraints on the DEP for all hops as

$$\max_{\psi \in \Psi, \{m_h, P_{T_h}^{(m_h)}, \Omega_{T_h}^{(m_h)}\}_{h \in \psi}} U(\psi) \quad (17)$$

$$\text{s.t. } P_{T_h}^{(m_h)} \leq P_{\max}, \quad \forall h \in \psi, \quad (18)$$

$$\Omega_{\text{lb}}^{(m_h)} \leq \Omega_{T_h}^{(m_h)} \leq \Omega_{\text{ub}}^{(m_h)}, \quad \forall h \in \psi, \quad (19)$$

$$\mathbb{P}_{\text{DEP},h} \geq \mathbb{P}_{\text{DEP, target}}, \quad \forall h \in \psi, \quad (20)$$

where $\mathbb{P}_{\text{DEP, target}}$ designates the target DEP for each of links in the selected route.

3) *Latency Minimization*: The third optimization problem is the *end-to-end latency minimization* with DEP requirements which can be expressed as

$$\min_{\psi \in \Psi, \{m_h, P_{T_h}^{(m_h)}, \Omega_{T_h}^{(m_h)}\}_{h \in \psi}} L(\psi) \quad (21)$$

$$\text{s.t. } P_{T_h}^{(m_h)} \leq P_{\max}, \quad \forall h \in \psi, \quad (22)$$

$$\Omega_{\text{lb}}^{(m_h)} \leq \Omega_{T_h}^{(m_h)} \leq \Omega_{\text{ub}}^{(m_h)}, \quad \forall h \in \psi, \quad (23)$$

$$\mathbb{P}_{\text{DEP},h} \geq \mathbb{P}_{\text{DEP, target}}, \quad \forall h \in \psi. \quad (24)$$

Note that, since the constraints on the target DEP in (20) and (24) is for each hop $h \in \psi$, when an end-to-end DEP requirement ϵ is needed, one can set $\mathbb{P}_{\text{DEP, target}} = \epsilon^{1/N_\psi}$ to satisfy $\mathbb{P}_{\text{DEP}}(\psi) = \prod_{h \in \psi} \mathbb{P}_{\text{DEP},h} \geq \epsilon$. The optimization with a constraint on the end-to-end DEP is an interesting topic and is left for future work.

In Section III, we will investigate the resource allocation problems for single-hop communication $h \in \psi$ where $\psi \in \Psi$. Then, in Section IV, we will develop optimal routing algorithms for the three optimization problems in this section based on the derived solutions in Section III.

III. SINGLE-HOP OPTIMIZATION

In this section, we develop novel joint power, bandwidth, and modality selection techniques for single-hop resource allocation problems. For any given single-hop link h , the resource allocation problem for the end-to-end DEP maximization problem is given by

$$\max_{m_h, P_{T_h}^{(m_h)}, \Omega_{T_h}^{(m_h)}} \mathbb{P}_{\text{DEP},h} \quad (25)$$

$$\text{s.t. } P_{T_h}^{(m_h)} \leq P_{\max}, \quad (26)$$

$$\Omega_{\text{lb}}^{(m_h)} \leq \Omega_{T_h}^{(m_h)} \leq \Omega_{\text{ub}}^{(m_h)}, \quad (27)$$

$$U_h \geq U_{\text{target}}. \quad (28)$$

Also, as the minimization of the latency L_h is equivalent to the maximization of the throughput U_h from (11), the single-hop

resource allocation problem for the throughput maximization and the latency minimization problem is formulated as

$$\max_{m_h, P_{T_h}^{(m_h)}, \Omega_{T_h}^{(m_h)}} U_h \quad (29)$$

$$\text{s.t. } P_{T_h}^{(m_h)} \leq P_{\max}, \quad (30)$$

$$\Omega_{\text{lb}}^{(m_h)} \leq \Omega_{T_h}^{(m_h)} \leq \Omega_{\text{ub}}^{(m_h)}, \quad (31)$$

$$\mathbb{P}_{\text{DEP},h} \geq \mathbb{P}_{\text{DEP},\text{target}}. \quad (32)$$

A. Single-hop DEP Maximization

Let us first examine the DEP maximization problem in (25). When a modality m_h is given, the problem in (25) becomes

$$\max_{P_{T_h}^{(m_h)}, \Omega_{T_h}^{(m_h)}} \mathbb{P}_{\text{DEP},h} \quad (33)$$

$$\text{s.t. } P_{T_h}^{(m_h)} \leq P_{\max}, \quad (34)$$

$$\Omega_{\text{lb}}^{(m_h)} \leq \Omega_{T_h}^{(m_h)} \leq \Omega_{\text{ub}}^{(m_h)}, \quad (35)$$

$$U_h \geq U_{\text{target}}. \quad (36)$$

To detect the communication over h , Willie obtains the optimal detection threshold δ_h that minimizes the DEP $\mathbb{P}_{\text{DEP},h}$ in (7) and (33). By differentiating $\mathbb{P}_{\text{DEP},h}$ with respect to δ_h , we have

$$\frac{\partial \mathbb{P}_{\text{DEP},h}}{\partial \delta_h} = \frac{\exp\left(-\frac{L\delta_h}{\Omega_{T_h}^{(m_h)} N_0}\right) L^L \delta_h^{L-1}}{\left(\Omega_{T_h}^{(m_h)} N_0\right)^L} - \frac{\exp\left(-\frac{L\delta_h}{P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2 + \Omega_{T_h}^{(m_h)} N_0}\right) L^L \delta_h^{L-1}}{\left(P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2 + \Omega_{T_h}^{(m_h)} N_0\right)^L}, \quad (37)$$

and hence the optimal δ_h that meets $\frac{\partial \mathbb{P}_{\text{DEP},h}}{\partial \delta_h} = 0$ is represented as

$$\delta_h = \frac{\left(P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2 + \Omega_{T_h}^{(m_h)} N_0\right) \Omega_{T_h}^{(m_h)} N_0}{P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2} \times \ln \left(\frac{P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2 + \Omega_{T_h}^{(m_h)} N_0}{\Omega_{T_h}^{(m_h)} N_0} \right). \quad (38)$$

Then, by substituting the threshold δ_h in (38) into the DEP $\mathbb{P}_{\text{DEP},h}$ in (7), we get

$$\begin{aligned} \mathbb{P}_{\text{DEP},h} &= 1 - \frac{1}{\Gamma(L)} \left(\gamma(L, A(X_h)) - \gamma(L, B(X_h)) \right) \quad (39) \\ &= 1 - \frac{1}{\Gamma(L)} \int_{B(X_h)}^{A(X_h)} t^{L-1} e^{-t} dt, \end{aligned}$$

where $A(x) \triangleq L(1+x) \ln(1+\frac{1}{x})$, $B(x) \triangleq Lx \ln(1+\frac{1}{x})$, and $X_h \triangleq \frac{\Omega_{T_h}^{(m_h)} N_0}{P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2}$ which is equivalent to the inverse of the signal-to-noise ratio (SNR) at Willie. In the following lemma, we derive that $\mathbb{P}_{\text{DEP},h}$ in (39) increases with X_h .

Lemma 1. *The DEP at Willie $\mathbb{P}_{\text{DEP},h}$ in (39) is an increasing function of $X_h \triangleq \frac{\Omega_{T_h}^{(m_h)} N_0}{P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2}$.*

Proof. The first and second derivatives of $A(x)$ and $B(x)$ with respect to x are respectively given by

$$\begin{aligned} \frac{\partial A(x)}{\partial x} &= L \left(\ln \left(1 + \frac{1}{x} \right) - \frac{1}{x} \right), \\ \frac{\partial^2 A(x)}{\partial x^2} &= \frac{L}{x^2(x+1)}, \\ \frac{\partial B(x)}{\partial x} &= L \left(\ln \left(1 + \frac{1}{x} \right) - \frac{1}{x+1} \right), \\ \frac{\partial^2 B(x)}{\partial x^2} &= -\frac{L}{x(x+1)^2}. \end{aligned}$$

As $\frac{\partial^2 A(x)}{\partial x^2} \geq 0$, $\frac{\partial A(x)}{\partial x}$ is an increasing function of x . In addition, $\frac{\partial A(x)}{\partial x} \leq 0$ since $\frac{\partial A(x)}{\partial x}$ converges to 0 when $x \rightarrow \infty$, and this leads to the fact that $A(x)$ is a decreasing function of x . On the other hand, $\frac{\partial B(x)}{\partial x}$ is decreased with x as $\frac{\partial^2 B(x)}{\partial x^2} \leq 0$. Here, $\frac{\partial B(x)}{\partial x} \geq 0$ due to the fact that $\frac{\partial B(x)}{\partial x}$ goes to 0 when $x \rightarrow \infty$, and therefore $B(x)$ is an increasing function of x . Based on these observations, it is shown that the range of the integration in (39) becomes smaller as X_h gets larger, and this concludes the proof. $\square$

Now, let us investigate the influence of the transmit power $P_{T_h}^{(m_h)}$ on U_h in (9) and $\mathbb{P}_{\text{DEP},h}$ in (39). For any given bandwidth $\Omega_{T_h}^{(m_h)}$, when $P_{T_h}^{(m_h)}$ increases, the throughput U_h grows and $\mathbb{P}_{\text{DEP},h}$ degrades since $\mathbb{P}_{\text{DEP},h}$ decays as X_h gets smaller. Thus, when $\Omega_{T_h}^{(m_h)}$ is given, in order to maximize the DEP $\mathbb{P}_{\text{DEP},h}$ under the constraint on the throughput in (36), the optimal transmit power $P_{T_h}^{(m_h)}$ should satisfy the constraint in (36) with equality and it can be expressed as

$$P_{T_h}^{(m_h)} = \frac{\Omega_{T_h}^{(m_h)} N_0}{|g_{T_h,R_h}^{(m_h)}|^2} \left(2^{U_{\text{target}}/\Omega_{T_h}^{(m_h)}} - 1 \right). \quad (40)$$

Next, let us look at the impact of the bandwidth $\Omega_{T_h}^{(m_h)}$ on the DEP $\mathbb{P}_{\text{DEP},h}$ in (39) when $P_{T_h}^{(m_h)}$ is computed as in (40). By differentiating the transmit power $P_{T_h}^{(m_h)}$ with respect to the bandwidth $\Omega_{T_h}^{(m_h)}$, we have

$$\begin{aligned} \frac{\partial P_{T_h}^{(m_h)}}{\partial \Omega_{T_h}^{(m_h)}} &= \frac{N_0}{|g_{T_h,R_h}^{(m_h)}|^2} \left(2^{U_{\text{target}}/\Omega_{T_h}^{(m_h)}} \left(1 - \frac{U_{\text{target}} \ln(2)}{\Omega_{T_h}^{(m_h)}} \right) - 1 \right), \\ \frac{\partial^2 P_{T_h}^{(m_h)}}{\partial \{\Omega_{T_h}^{(m_h)}\}^2} &= \frac{N_0 (U_{\text{target}} \ln(2))^2}{|g_{T_h,R_h}^{(m_h)}|^2 \left(\Omega_{T_h}^{(m_h)} \right)^3} 2^{U_{\text{target}}/\Omega_{T_h}^{(m_h)}}. \end{aligned}$$

As $\frac{\partial^2 P_{T_h}^{(m_h)}}{\partial \{\Omega_{T_h}^{(m_h)}\}^2} \geq 0$ and $\frac{\partial P_{T_h}^{(m_h)}}{\partial \Omega_{T_h}^{(m_h)}}$ goes to 0 when $\Omega_{T_h}^{(m_h)} \rightarrow \infty$, $\frac{\partial P_{T_h}^{(m_h)}}{\partial \Omega_{T_h}^{(m_h)}} \leq 0$, and this means that the optimal transmit power $P_{T_h}^{(m_h)}$ decreases with the bandwidth $\Omega_{T_h}^{(m_h)}$. In this sense, from the definition of $X_h \triangleq \frac{\Omega_{T_h}^{(m_h)} N_0}{P_{T_h}^{(m_h)} |g_{T_h,W}^{(m_h)}|^2}$ and Lemma 1,

the DEP $\mathbb{P}_{\text{DEP},h}$ is an increasing function of the bandwidth $\Omega_{T_h}^{(m_h)}$ with the optimal transmit power $P_{T_h}^{(m_h)}$ in (40). Hence, the optimal bandwidth $\Omega_{T_h}^{(m_h)}$ is

$$\Omega_{T_h}^{(m_h)} = \Omega_{\text{ub}}^{(m_h)}, \quad (41)$$

and the corresponding optimal transmit power $P_{T_h}^{(m_h)}$ becomes

$$P_{T_h}^{(m_h)} = \frac{\Omega_{\text{ub}}^{(m_h)} N_0}{|g_{T_h, R_h}^{(m_h)}|^2} \left(2^{U_{\text{target}}/\Omega_{\text{ub}}^{(m_h)}} - 1 \right). \quad (42)$$

If $P_{T_h}^{(m_h)}$ in (42) is higher than P_{max} , the throughput requirement in (36) cannot be fulfilled, and therefore the problem in (33) is not feasible. It should be noted that this is another motivation for developing multi-hop multi-modal routing algorithms that find routes from a source to a destination when the single link between the source and destination with any of the available modalities does not meet the requirements.

Until now, we have identified the optimal transmit power $P_{T_h}^{(m_h)}$ and bandwidth $\Omega_{T_h}^{(m_h)}$ for any given modality. Note that due to the difference in the center frequencies for the various modalities, the characteristics of the channels from Alice to Bob and Willie vary over modalities even when the locations of Alice, Bob and Willies are fixed. Thus, we provide a modality selection approach that chooses one modality that exhibits the highest DEP at Willie among M available modalities.

Leveraging the solutions in (41) and (42), X_h in (39) is represented as

$$X_h = \frac{|g_{T_h, R_h}^{(m_h)}|^2}{|g_{T_h, W}^{(m_h)}|^2 \left(2^{U_{\text{target}}/\Omega_{\text{ub}}^{(m_h)}} - 1 \right)}. \quad (43)$$

As shown in Lemma 1, $\mathbb{P}_{\text{DEP},h}$ is an increasing function X_h for $\forall h \in \psi$, and hence the modality m_h having the biggest DEP can be chosen as

$$m_h = \arg \max_{m=1, \dots, M} \frac{|g_{T_h, R_h}^{(m)}|^2}{|g_{T_h, W}^{(m)}|^2 \left(2^{U_{\text{target}}/\Omega_{\text{ub}}^{(m)}} - 1 \right)}. \quad (44)$$

Note that the modality selection method in (44) chooses the DEP maximizing modality only based on the channel gains and allocated bandwidths without computing the complicated DEPs for different modalities in (7).

The proposed algorithm for the DEP maximization of single-hop communications is summarized as below.

Algorithm 1: Single-hop DEP Maximization

1. Identify the optimal modality m_h which provides the highest DEP following (44).
 2. Obtain the optimal bandwidth $\Omega_{T_h}^{(m_h)}$ as in (41).
 3. Calculate the optimal transmit power $P_{T_h}^{(m_h)}$ as in (42).
-

B. Single-hop Throughput Maximization and Latency Minimization

Now, we concentrate on the throughput maximization problem in (29) for a given modality m_h . Let us first examine the optimization of bandwidth $\Omega_{T_h}^{(m_h)}$. From Lemma 1, the DEP $\mathbb{P}_{\text{DEP},h}$ gets larger when $\Omega_{T_h}^{(m_h)}$ grows. Also, by differentiating U_h in (9) with respect to $\Omega_{T_h}^{(m_h)}$, we have

$$\begin{aligned} \frac{\partial U_h}{\partial \Omega_{T_h}^{(m_h)}} &= \frac{1}{\ln(2)} \\ &\times \left[\ln \left(1 + \frac{P_{T_h}^{(m_h)} |g_{T_h, R_h}^{(m_h)}|^2}{\Omega_{T_h}^{(m_h)} N_0} \right) - \frac{P_{T_h}^{(m_h)} |g_{T_h, R_h}^{(m_h)}|^2}{P_{T_h}^{(m_h)} |g_{T_h, R_h}^{(m_h)}|^2 + \Omega_{T_h}^{(m_h)} N_0} \right], \end{aligned} \quad (45)$$

and after some computations, one can readily show that $\frac{\partial U_h}{\partial \Omega_{T_h}^{(m_h)}}$ is always positive. Then, since both the DEP $\mathbb{P}_{\text{DEP},h}$ and the throughput U_h are increasing functions of $\Omega_{T_h}^{(m_h)}$, the optimal $\Omega_{T_h}^{(m_h)}$ is given by

$$\Omega_{T_h}^{(m_h)} = \Omega_{\text{ub}}^{(m_h)}. \quad (46)$$

On the other hand, when the transmit power $P_{T_h}^{(m_h)}$ becomes higher, the throughput U_h increases and the DEP $\mathbb{P}_{\text{DEP},h}$ decays. In this sense, to maximize U_h while satisfying the DEP requirement in (32), the optimal transmit power $P_{T_h}^{(m_h)}$ should meet the DEP constraint with equality. As the DEP $\mathbb{P}_{\text{DEP},h}$ is a monotonically decreasing function of $P_{T_h}^{(m_h)}$ as shown in Lemma 1, the optimal $P_{T_h}^{(m_h)}$ that fulfills $\mathbb{P}_{\text{DEP},h} = \mathbb{P}_{\text{DEP, target}}$ can be found by utilizing one-dimensional line search method such as the *bisection method* [33].

Lastly, after evaluating the throughput for each of the multiple modalities based on the derived solutions for the bandwidth and power, the best modality exhibiting the highest throughput is selected as

$$m_h = \arg \max_{m=1, \dots, M} U_h. \quad (47)$$

The proposed strategy for the throughput maximization and latency minimization of single-hop communications is summarized as follows:

Algorithm 2: Single-hop Throughput Maximization and Latency Minimization

1. Compute the optimal bandwidth $\Omega_{T_h}^{(m_h)}$ as in (46).
 2. Find the optimal transmit power $P_{T_h}^{(m_h)}$ that satisfies $\mathbb{P}_{\text{DEP},h} = \mathbb{P}_{\text{DEP, target}}$ using the bisection method.
 3. Identify the optimal modality m_h as in (47).
-

So far, we have introduced the optimal modality, transmit power, and bandwidth allocation techniques for single-hop optimization problems. In the next section, we will investigate multi-hop routing algorithms building on the single-hop optimization described in this section for the three optimization problems in Section II-D.

IV. MULTI-HOP ROUTING TECHNIQUES

In this section, we provide the optimal multi-hop routing techniques based on the developed optimal resource allocation methods for single-hop communications in Section III.

A. End-to-end DEP Maximization

Let us first look at the end-to-end DEP maximization problem in (13) when a route ψ is given. As defined in (8), for any given route ψ , the end-to-end DEP $P_{\text{DEP}}(\psi)$ is the multiplication of the DEPs of the hops along the route $\{P_{\text{DEP},h}\}_{h \in \psi}$. Also, the requirement on the end-to-end throughput is satisfied when the throughput of each hop fulfills the requirement, i.e., $U_h \geq U_{\text{target}}$ for $h \in \psi$. Therefore, for any given route $\psi = (h_1, \dots, h_{N_\psi})$, the joint N_ψ -hop optimization can be decomposed into independent N_ψ single-hop optimization problems where each problem is given by (25). Here, the single-hop DEP maximization problems can be solved with Algorithm 1.

Now, let us focus on the route selection for the end-to-end DEP maximization. With the optimized modality, transmit power, and bandwidth, we express the route selection problem as

$$\max_{\psi \in \Psi} P_{\text{DEP}}(\psi). \quad (48)$$

Based on the definition of $P_{\text{DEP}}(\psi)$ in (8), without loss of generality, the problem in (48) is reformulated as

$$\begin{aligned} \max_{\psi \in \Psi} \prod_{h \in \psi} P_{\text{DEP},h} &\Leftrightarrow \max_{\psi \in \Psi} \sum_{h \in \psi} \ln(P_{\text{DEP},h}) \\ &\Leftrightarrow \min_{\psi \in \Psi} \sum_{h \in \psi} \ln\left(\frac{1}{P_{\text{DEP},h}}\right). \end{aligned} \quad (49)$$

The objective function of the reformulated problem in (49) is the sum of $\ln\left(\frac{1}{P_{\text{DEP},h}}\right)$ for all hops. Thus, by modeling the HetNet as a graph and defining the edge weight of link h as $\ln\left(\frac{1}{P_{\text{DEP},h}}\right)$, the end-to-end DEP maximizing routing problem becomes equivalent to the shortest-path problem. Here, the graph is a directed graph since the DEP at Willie for the transmission from A to B may be different from that for the communication from B to A as the channels to Willie from A and B may not be the same. The shortest-path problem for the directed graph can be efficiently solved by Dijkstra's algorithm [34].

The joint route and resource allocation method for the end-to-end DEP maximization problem in (13) is summarized as follows:

Algorithm 3: End-to-end DEP Maximization

1. Obtain the optimal modality, transmit power, and bandwidth for all possible single-hop links from Algorithm 1.
 2. Define the edge weight by $\ln\left(\frac{1}{P_{\text{DEP},h}}\right)$ for all links where $P_{\text{DEP},h}$ is the DEP with the optimal resource allocation from Algorithm 1 for link h .
 3. Identify the optimal route of the problem in (49) using Dijkstra's algorithm.
-

B. End-to-end Throughput Maximization

In this subsection, we examine the end-to-end throughput maximization in (17). For any given route ψ , since the throughputs of all hops along the route are independent, the end-to-end throughput $U(\psi)$ in (10), which is the minimum of the throughputs, is maximized when the throughput of each hop is maximized. In addition, we can see that the constraints in (17) are decoupled over hops. Hence, for any given route $\psi = (h_1, \dots, h_{N_\psi})$, without loss of generality, the joint multi-hop resource allocation problem is decomposed into multiple single-hop throughput maximization problems in (29). The optimal solution for the problem in (29) can be obtained with Algorithm 2.

With the optimized modality, transmit power, and bandwidth, the end-to-end throughput maximizing routing problem is represented as

$$\max_{\psi \in \Psi} \min_{h \in \psi} U_h. \quad (50)$$

As the aim of this problem is to find the route from Alice to Bob that maximizes the minimum of the throughputs along the route, the problem in (50) is equivalent to the widest path problem which is also known as the maximum capacity path problem.

Due to the DEP constraint in (20), the optimized throughput for the single-hop communication from A to B are not the same as that for the link from B to A as the channels to Willie from A and B may be different. Therefore, by modeling the HetNet as a directed graph with edge weights that are equal to the throughputs $\{U_h\}$, one can identify the optimal route for the end-to-end throughput maximization problem in (50) by solving the widest path problem [34].

The proposed joint optimal route and resource allocation scheme for the end-to-end throughput optimization in (17) is summarized as below.

Algorithm 4: End-to-end Throughput Maximization

1. Calculate the optimal modality, transmit power, and bandwidth to maximize throughput for all possible single-hop links using Algorithm 2.
 2. Define the edge weight by U_h for all links where U_h is the throughput with the optimal resource allocation from Algorithm 2 for link h .
 3. Determine the optimal route of the problem in (50) by solving the widest path problem.
-

C. End-to-end Latency Minimization

In this subsection, we focus on the end-to-end latency minimization problem in (21). When a route $\psi \in \Psi$ is given, the end-to-end latency is the sum of the latencies of the hops in the route, and the constraints in (21) for multiple hops are not correlated. Thus, for any given route ψ , the multi-hop joint resource allocation problem for the end-to-end latency minimization is decomposed into multiple single-hop latency minimization problems which are equivalent to the problem in (29) owing to the definition of the latency $L_h = D/U_h$

in (11). The optimal modality, transmit power, and bandwidth allocation for the single-hop problems can be derived with Algorithm 2.

Then, the end-to-end latency minimizing route selection problem is formulated as

$$\min_{\psi \in \Psi} \sum_{h \in \psi} L_h, \quad (51)$$

and this problem becomes the shortest path problem when we model the HetNet as a directed graph and define the edge weight for each single-hop link as the latency of the link with the optimized modality, transmit power, and bandwidth. The optimal route of the shortest path problem can be found by Dijkstra's algorithm [34].

The developed end-to-end latency minimizing joint optimization technique is summarized as below.

Algorithm 5: End-to-end Latency Minimization

1. Compute the optimal modality, transmit power, and bandwidth for all possible single-hop links from Algorithm 2.
 2. Define the edge weight by L_h for all links where L_h is the latency with the optimal resource allocation from Algorithm 2 for link h .
 3. Identify the optimal route of the problem in (51) by leveraging the Dijkstra's algorithm.
-

V. NUMERICAL RESULTS

In this section, we provide numerical simulation results to validate the performance of the proposed joint routing and resource allocation strategies based on a 3D simulation environment ($250 \times 250 \times 9.5$) m³ as illustrated in Fig. 2 with 36 nodes (shown as the red cones) where vertically polarized short dipole antennas on each node are positioned at a height of 3 m above ground. Here, the green cuboids in Fig. 2 indicate concrete buildings. We consider three communication modalities M_1 , M_2 , and M_3 whose center frequencies are 400 MHz, 900 MHz, and 2.4 GHz, respectively. To obtain channel data, ray-tracing approaches are utilized using a commercial software [30]. For each of the 36 nodes and each of the three modalities, the received signal is computed at a densely sampled grid of receivers with a spacing of 1 m (resulting in a total of 62,500 receiver locations per node per modality). An example of one of the simulation results when node 28 is a transmitting node is given in Fig. 3 where the received channel gains for the three modalities are plotted. After generating the channel gain data, we select values for other required parameters. Here, unless otherwise stated, we assume that $L = 100$, $P_{\max} = 10$ dBm, $N_0 = -110$ dBm, $\Omega_{\text{lb}}^{(m)} = 0$ MHz, $\Omega_{\text{ub}}^{(m)} = 4$ MHz, and $\mathbb{P}_{\text{DEP, target}} = 0.9$.

First, we present the results for the single-hop multi-modal communication performance described in Section III. Fig. 4 presents the DEP, throughput, and latency of the single-hop communication between Alice (node 29) and Bob (node 9) as a function of Y coordinate of Willie location $L_{W,y}$ where Willie is located at $(50, L_{W,y})$, $U_{\text{target}} = 2$ Mbps, and $D = 1$ Mb. A

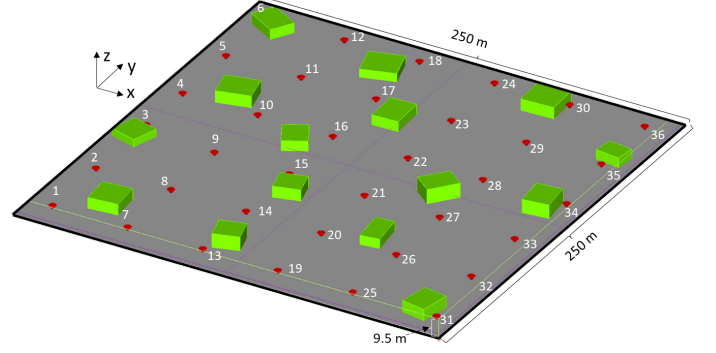


Fig. 2. A 36-node network topology deployed in a multi-building scenario simulated using ray tracing [30]. Here, the red markers are the nodes; the green cuboids are the concrete building models; and the grey region is the ground. For each node, a separate simulation is run to acquire the channel data between the node and a densely sampled grid of receivers each of them separated by 1 m (i.e., 62,500 receivers).

key observation is that each of the three modalities exhibits a different performance due to the difference in channels among Alice, Bob, and Willie. Another observation is that none of the three modalities always provides better performance than others for various locations of Willie. This implies that it is important to adaptively select the best modality among the available modalities based on the channel condition. When $60 \leq L_{W,y} \leq 110$ or $145 \leq L_{W,y} \leq 180$, as observed in Fig. 3, the received channel gain at Willie from Alice is high, and therefore the resultant DEP and throughput become smaller, and the latency gets larger as shown in Fig. 4.

In Fig. 5, we provide the selected modality for the DEP maximization of the single-hop communication between Alice and Bob for various locations of Willie where Bob is node 15 and Alice is node 28 or 2. Here, Alice chooses the optimal communication modality M_1 , M_2 , or M_3 to communicate with Bob. The optimal modalities for all possible locations of Willie are highlighted in three different colors. As an example, the areas A, B, and C signify the selection of M_1 , M_2 , and M_3 , respectively. These modality selection maps suggest the need for developing a modality selection approach that judiciously selects one modality based on the wireless channel and adapts to the dynamic nature of the wireless environment. It should be noted that the more diverse and higher number of modalities could be exploited to further enhance the overall performance. Note that the DEP performance is influenced by not only the channel from Alice to Willie, but also the channel from Alice to Bob due to the constraint on the throughput at Bob. Thus, the modality selection map will change as the position of Bob varies even if the locations of Alice and Willie remain the same.

Fig. 6 demonstrates the end-to-end DEP, throughput, and latency of the proposed optimal joint route and resource allocation techniques for the multi-hop multi-modal routing (as described in Section IV) from Alice (node 1) to Bob (node 36) considering different $L_{W,x}$ where the location of Willie is $(L_{W,x}, 125)$, $U_{\text{target}} = 5$ Mbps, and $D = 10$ Mb. A key observation is that the end-to-end performances are highly variable function of the location of Willie since the detection

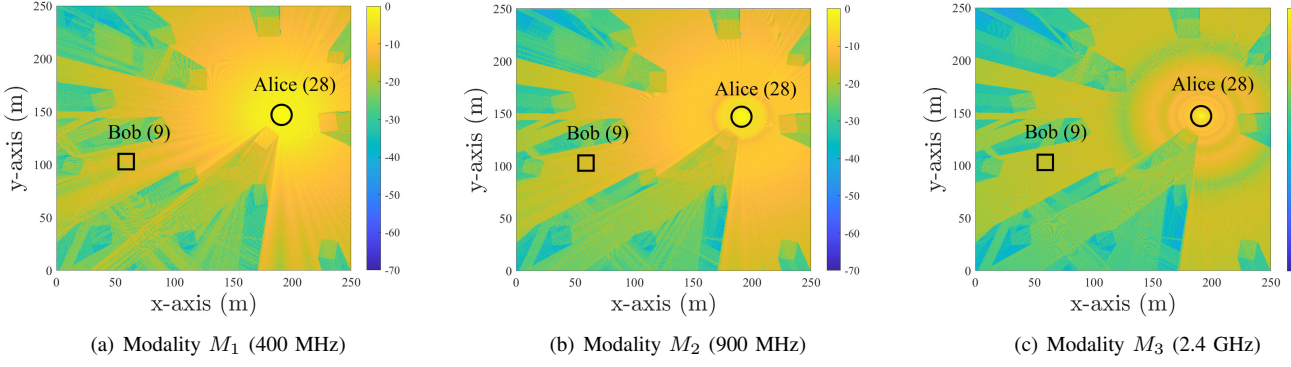


Fig. 3. Channel gains of the three modalities in dB when node 28 is used as the transmitter.

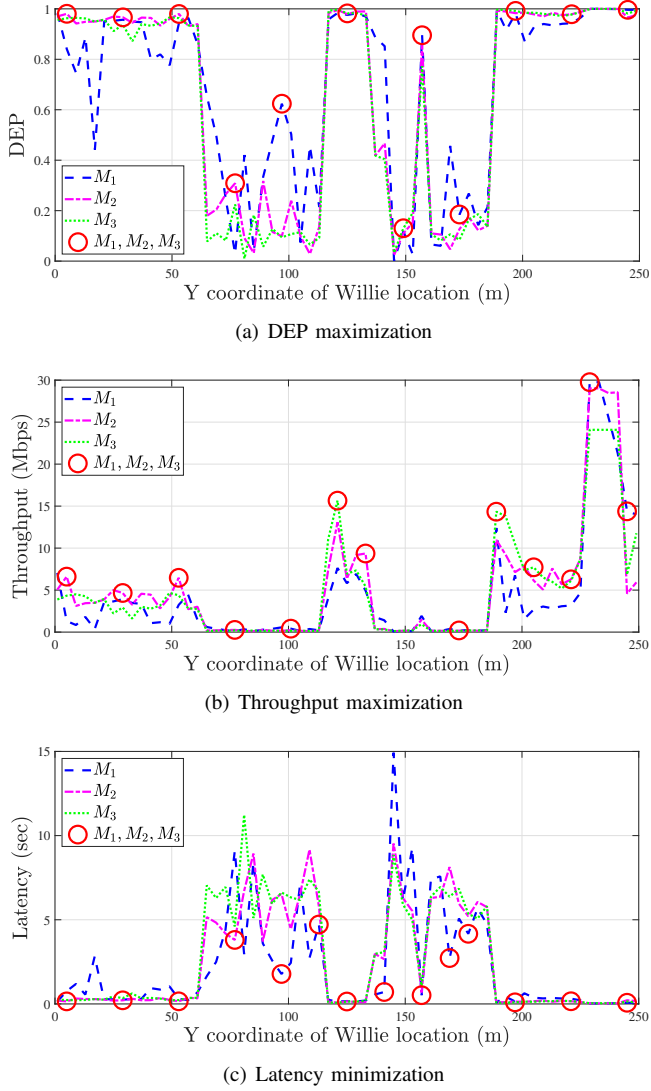


Fig. 4. Performances of the proposed algorithms for the three single-hop resource allocation problems.

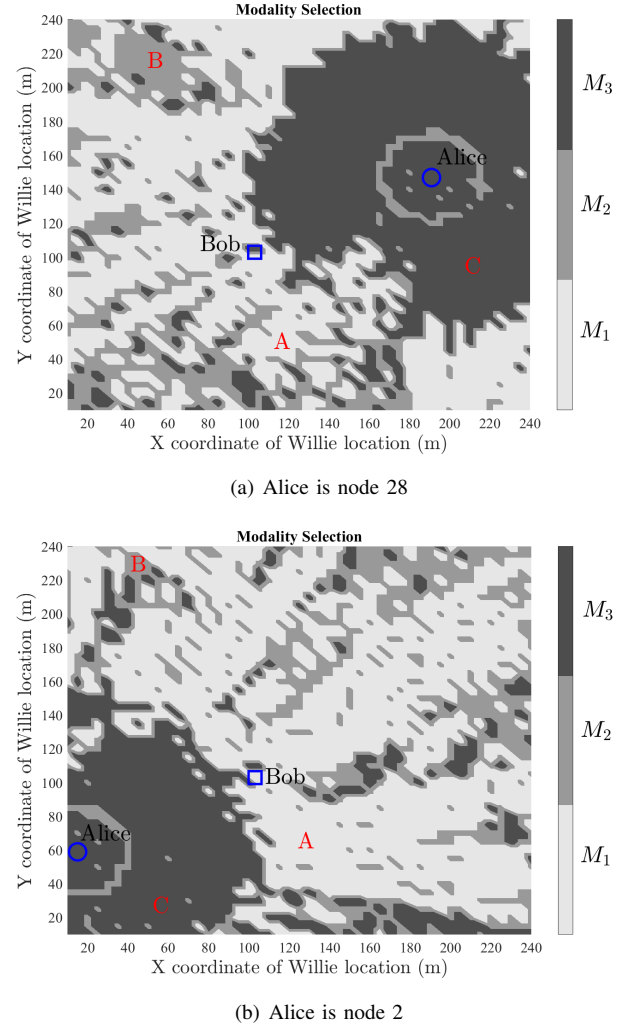


Fig. 5. The selected modality for the communication from Alice to Bob as a function of the location of Willie.

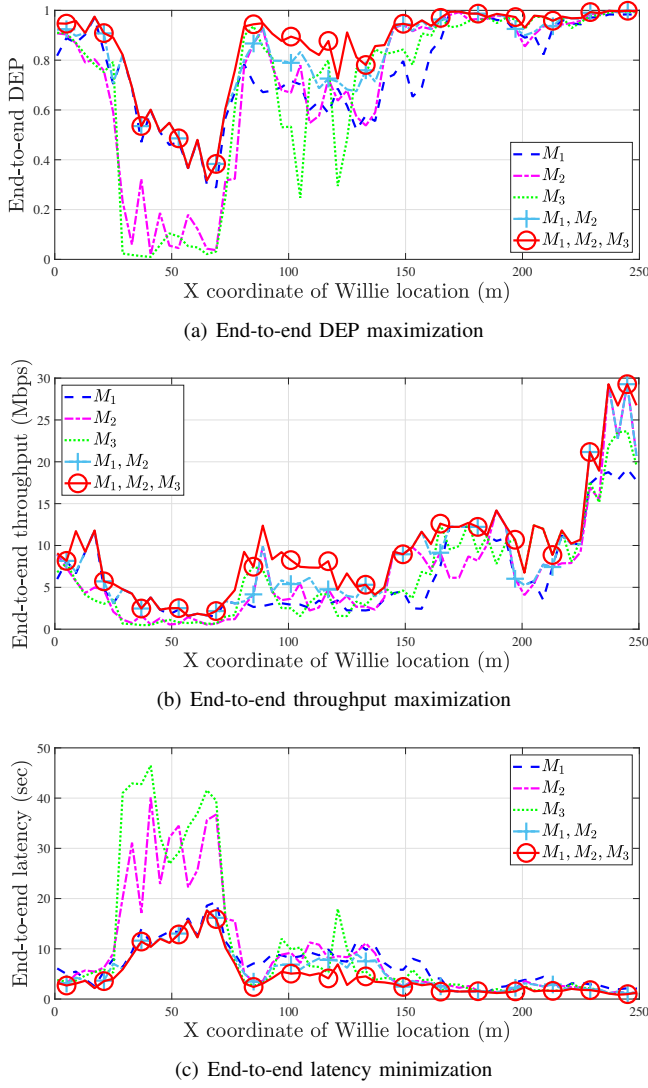


Fig. 6. Performances of the proposed algorithms for the three multi-hop joint route and resource allocation problems.

performance at Willie is determined by the received channel gain at Willie from Alice. In addition, since each modality experiences significantly different channels from Alice to Bob and Willie, the three modalities exhibit different performances, and none of the three is superior to the others. Note that, for single-hop communications, the modality selection provides the best of the performances of the three modalities. On the other hand, for multi-hop communications, each transmitter in the route selects its own communication modality, and hence the selected modalities along the route may not be the same. By selecting the most optimal modality for each hop in a route, we can achieve the performance in ways that cannot be attained with one of the three modalities. To our knowledge, this approach is fundamentally different from conventional HetNets where a node can switch between multiple independent networks (e.g., WiFi and 4G/5G on cellular phones). One observation is that, when $25 \leq L_{W,x} \leq 90$, in order to ensure the covertness of the communication, Alice may not be able to send data to nodes located on the top side and prefers to

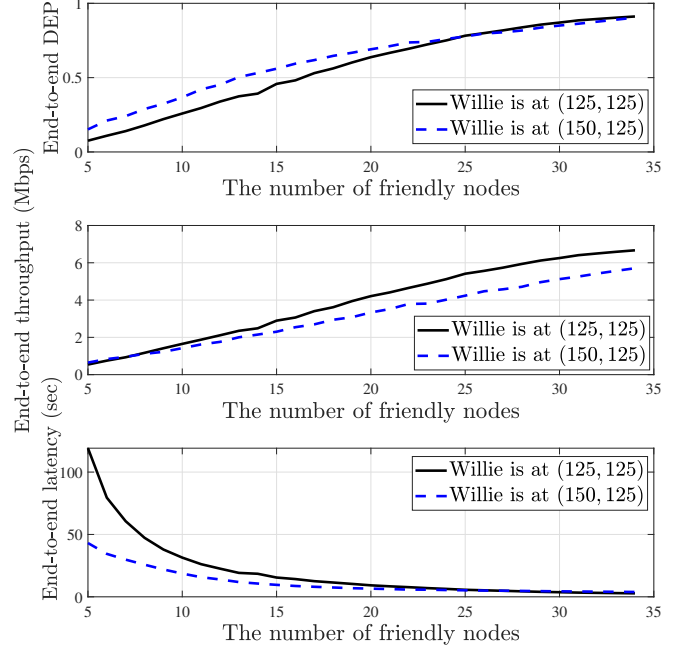


Fig. 7. The end-to-end performances as a function of the number of friendly nodes for the communication from Alice (node 1) to Bob (node 36).

communicate with nodes on the right side (since Alice is node 1 in this case and Willie's y-coordinate is 125 m.) This leads to a decreased number of candidates for the next hop at Alice. Therefore, as observed in Fig. 6, when $25 \leq L_{W,x} \leq 90$, the end-to-end DEP and throughput become smaller, and the end-to-end latency gets higher.

In Fig. 7, we present the end-to-end performances as a function of the number of friendly nodes for the communication between Alice (node 1) and Bob (node 36) considering two different locations of Willie, i.e., Willie is located at (125, 125) or (150, 125), when $U_{\text{target}} = 5$ Mbps, and $D = 10$ Mb. Here, the friendly nodes are the nodes supporting the communication from Alice to Bob, and some of the friendly nodes will be selected as relaying nodes. For any given number of friendly nodes N_F , we randomly select N_F nodes from 34 nodes and evaluate the end-to-end performances with the chosen nodes. Then, we repeat the random selection for 2,500 realizations to compute the average end-to-end performances. From Fig. 7, we can observe that the end-to-end performances are improved with the number of friendly nodes. Also, it is shown that the performances vary significantly as a function of Willie's location, and DEP and latency for the two different locations of Willie become similar when the number of relaying nodes is large. Another observation is that the DEP and latency tend to be saturated as the number of friendly nodes gets higher, and on the other hand, the throughput tends to keep increasing with the number of friendly nodes.

Fig. 8 illustrates the optimal routes and selected modalities along the route from Alice (node 1) to Bob (node 36) for the three performance metrics and for various locations of Willie

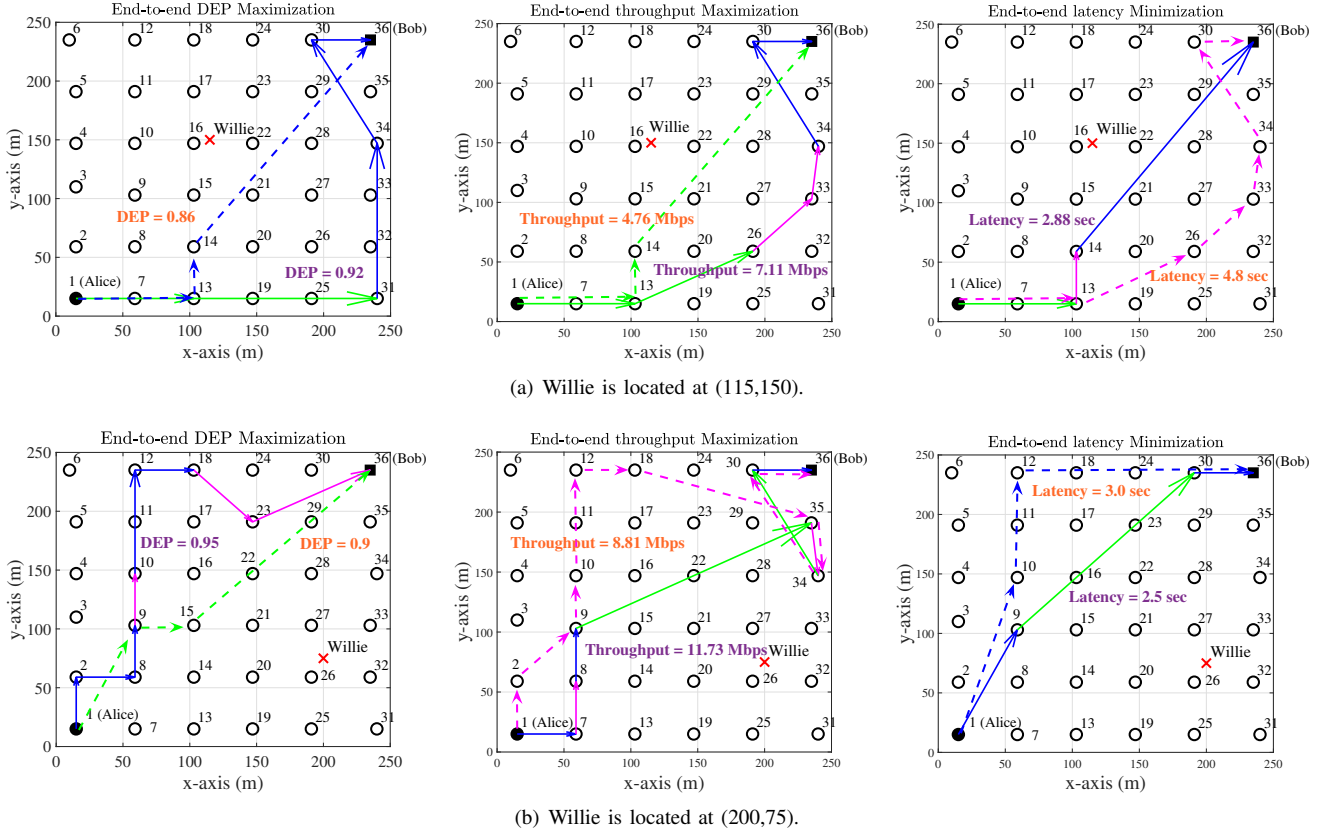


Fig. 8. Optimal routes and selected modalities for the three multi-hop joint route and resource allocation problems. The hops with modalities M_1 , M_2 , and M_3 are highlighted in blue, magenta, and green, respectively. The routes with solid lines are the optimal routes with the three modalities, and the routes with dashed lines are the optimal routes with one modality. The end-to-end performances with three modalities and single modalities are highlighted in purple and orange, respectively.

when $U_{\text{target}} = 5$ Mbps and $D = 10$ Mb. From Fig. 8, it is shown that the optimal routes and chosen modalities at all nodes along the routes vary significantly with the location of Willie and the end-to-end performance metric. Another observation is that the established routes tend to avoid using relay nodes that are close to Willie. This is due to the need for increasing the DEP at Willie (i.e., for the end-to-end DEP maximization problem), or satisfy the requirement on the DEP for the end-to-end throughput maximization and end-to-end latency minimization problems. This behavior is especially pronounced for the end-to-end DEP maximization problem since the primary goal is to maximize the end-to-end DEP instead of fulfilling a constraint on the DEP. The optimal routes for this case tend to go around Willie ensuring the distance from Willie to any of the relay nodes in the route is large. It should be noted that this effect is not as pronounced for the end-to-end throughput maximization and end-to-end latency minimization problems. For comparison, we also include the selected routes when only one of the three modalities is used for all hops in the route in Fig. 8. By comparing the end-to-end performance with the three modalities to that with one modality, we can confirm that the end-to-end performance is enhanced by utilizing the available modalities efficiently.

VI. CONCLUSION

In this paper, we have investigated covert routing in a HetNet where Alice transmits confidential data to Bob with the aid of multiple relaying nodes where each transmitting node judiciously selects one modality among multiple available communication modalities. We have studied joint optimizations of route and resource allocation problems considering three performance metrics, the end-to-end DEP, end-to-end throughput, and end-to-end latency. First, for single-hop communications between Alice and Bob, we have proposed closed-form optimal transmit power, bandwidth, and modality allocation techniques for the three performance metrics. Next, for multi-hop multi-modal routing, we have developed the optimal joint route and resource allocation strategies by modeling the HetNet as directional graphs with edge weights that are defined based on the objectives of the problems. Based on extensive numerical simulations and simulated high-fidelity channel data for three different communication modalities, we have validated that the proposed optimal covert routing algorithms for the HetNet can enhance the performances of the HetNet by adaptively choosing the best modalities at all nodes in routes compared to the networks with one communication modality, or conventional HetNets that consist of multiple independent networks.

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